PROJECT REPORT

NSE Paper-Trading & Intraday Research System

Product vision • requirements • architecture • phased delivery plan

Field	Value
Status	Planning / prototype stage
Initial universe	Nifty 50 cash-equity constituents
Execution	Paper simulation only; no real-order submission
Virtual capital assumption	₹5,000 (confirm before final implementation)
Platform	Local Python application on Windows
Report version/date	v1.0 — 25 October 2026

1. Executive Summary

The project is to build a local Python-based intraday research and paper-trading application for Indian NSE cash-market stocks. It is intended for learning and evaluation, not automated live trading.

The first release scans the Nifty 50 universe, uses five-minute OHLCV candles, evaluates a precisely defined Opening Range Breakout (ORB) strategy, applies risk controls, simulates trades, backtests historical data, and displays results in a local dashboard.

It is intended for learning and evaluation, not automated live trading.

The existing repository at C:\Users\ankit\day-trader-skill is a US-focused Claude skill package. Its symbols, market times, and data assumptions should not be copied directly into NSE logic.

The agreed approach is to preserve the original modules and build NSE functionality separately under nse\. A Python test file currently runs successfully, but the full scanner and risk engine are not yet implemented.

2. Product Definition

2.1 Purpose and user

Primary user: one person running the system locally on a Windows computer.

Initial market: NSE cash equities; first universe is Nifty 50 constituents.

Initial timeframe: five-minute candles; daily candles may be used for context.

Proposed data source: Angel One SmartAPI, subject to account access, current documentation, API terms and entitlement checks.

Cost goal: local-first, no paid AI API required; ordinary Python calculations are sufficient.

Primary outcome: an auditable research tool that explains signals, risk, simulated fills and results.

2.2 Explicit non-goals

No real-order placement, modification, or broker execution in any phase.

No options, futures, leverage, margin, short selling, or crypto in the first release.

No claim that a strategy is 100% successful or guaranteed to profit.

No public cloud deployment or unattended live trading in the first release.

No dependence on an AI API for indicators, signals, risk checks or execution simulation.

3. Current Baseline and Constraints

Area	Known baseline / implication
Repository	day-trader-skill; original project is US-focused. Preserve it and build NSE code separately.
Current NSE files	nse__init__.py, nse\scanner__init__.py and nse\scanner\scanner.py.
Current scanner	scanner.py has only a print test; it is not yet a functional trading scanner.
Python	Python is installed and has successfully run the module; confirm active version before selecting dependency versions.
Computer	Previously reported Windows laptop with 16 GB RAM and integrated graphics should be adequate for 50-symbol scanning and moderate backtests.
Market data	Validate SmartAPI access, symbol-token mapping, historical data limits, permissions and request-rate limits.
Capital	₹5,000 is an initial virtual-capital assumption; confirm before final

4. Product Workflow

Load the dated Nifty 50 constituent list and validate the symbols and instrument tokens.

Check the NSE market calendar and determine whether the regular session is open.

Fetch or load historical/live market data through a provider interface.

Validate candle timestamps, OHLC values, volume, gaps, duplicates and freshness.

Build the opening range from the first three completed five-minute bars (09:15–09:30 IST).

Evaluate only completed candles against the documented ORB rules.

Send any candidate signal through the risk engine; block it if data, sizing or risk checks fail.

Simulate eligible trades using explicit fill, fee and slippage assumptions.

Record positions, cash, P&L, risk verdicts and all relevant timestamps.

Show scanner, portfolio, risk and backtest results on the local dashboard.

5. Scope by Phase

Phase	Scope	Exit criteria
0 — Foundation	Confirm requirements, virtual capital, Python environment, repository baseline and paper-only boundary.	Approved scope and validated project structure.
1 — Scanner core	Synthetic candles, ORB calculations, volume filter, candle validation and console output.	Unit tests cover valid and invalid signal cases.
2 — Universe	Dated official Nifty 50 constituents and validated symbol-token mapping.	50 unique expected members; unknown tokens are reported, never guessed.
3 — Market data	Provider interface, local cache, Angel One historical candles and optional streaming feed.	Freshness, retry, throttling and data-validation tests pass.
4 — Risk and paper ledger	Position sizing, stop/target checks, exposure/cash limits, daily circuit breaker, simulated fills and local ledger.	Risk tests pass; no broker order API is present.
5 — Backtesting	Chronological replay, fees, slippage, metrics and out-of-sample evaluation.	Reproducible trade log; no look-ahead bias.
6 — Dashboard	Local views for status, signals, risk, portfolio, journal and backtest.	Paper-only banner, stale-data warnings and export work.
7 — Paper observation	Run the simulator, reconcile data and logs, fix operational defects.	No critical defects; review evidence across 20–30 trading sessions.
8 — Maintenance	API/constituent updates, backups, regression tests and controlled expansion.	Versioned releases, backup/restore test and change log.

6. Functional Requirements

ID	Requirement	Details
FR-01	Universe management	Use a dated Nifty 50 constituent file; record effective date and handle constituent changes.
FR-02	Instrument mapping	Map symbols to verified NSE tokens from a validated instrument master; never guess.
FR-03	Market calendar	Use Asia/Kolkata timezone, exchange holidays and special sessions; never assume every weekday is open.
FR-04	Candle ingestion	Read fixtures, CSV, cached data, historical API and later a live feed through one interface.
FR-05	Data validation	Reject malformed OHLC, negative volume, duplicates, gaps, stale or incomplete bars; log the reason.
FR-06	Strategy evaluation	Evaluate completed bars and emit an explainable structured signal.
FR-07	Risk checks	Check quantity, stop, target, cash, exposure, daily loss, duplicate signals and trade limits.
FR-08	Paper execution	Simulate fills and costs; never call broker order endpoints.
FR-09	Portfolio ledger	Track cash, positions, open risk, realized/unrealized P&L, fees and trade history.
FR-10	Backtesting	Replay chronologically with costs, slippage and no look-ahead bias.
FR-11	Dashboard	Display market status, data freshness, signals, portfolio, risk and system health.
FR-12	Audit trail	Persist timestamps, strategy/config version, risk verdicts, fills and errors.
FR-13	Fail-safe	Invalid config, stale data or corrupted state blocks new simulated entries.
FR-14	Exports	Export signals, trades and backtest reports to CSV/JSON.

7. Non-Functional Requirements

Safety: no real-order submission function or broker order endpoint in the application.

Reliability: same inputs, code version and config should produce deterministic outputs.

Data integrity: timestamps must be normalized consistently to Asia/Kolkata; no silent data imputation for signals.

Performance: scan 50 symbols after each completed five-minute bar on the local computer without a GPU.

Security: secrets only in local environment variables or an approved secret store; never in chat, source files or logs.

Observability: structured logs, last-success timestamp, feed health and readable errors.

Maintainability: separate data, strategy, risk, paper execution, storage and UI modules.

Reproducibility: save configuration snapshots and versions with backtests and paper sessions.

Cost: no paid AI dependency; verify market-data access and terms before use.

8. Initial Strategy Specification — ORB Candidate

The ORB is a research candidate, not a validated edge. Freeze the rules before running a performance test.

Item	Initial proposed rule	Validation required
Universe	Nifty 50 NSE cash equities.	Use dated constituents; assess survivorship bias in historical research.
Bars	Five-minute OHLCV.	Verify provider timestamp convention and bar completion.
Opening range	Regular session 09:15 through 09:30 IST; high/low from first three five-minute bars.	Exclude pre-open data and define bar boundaries exactly.
Long trigger	A completed candle closes above opening-range high.	Compare single close, retest and other filters without hindsight.
Volume filter	Breakout volume exceeds configurable multiple of opening-range average volume.	Test robustness; three opening bars give a small baseline sample.
Entry fill	Conservative next-bar-open or documented breakout fill model.	Do not assume a price was available if it was not.
Stop	Below opening-range low for the first long-only candidate.	Validate distance and quantity; never move it after observing future data.
Target	Initial candidate 1.5 times initial risk.	Test after fees and slippage; ratio alone does not prove expectancy.
Entry cutoff	Configurable within regular session.	Finalize and test; no entries after cutoff.
Position handling	Long-only; one position per symbol; no averaging down.	Define same-bar stop/target conflict conservatively.

The earlier prototype is only a starter. It does not yet validate sessions, completed-bar status, quantity, account-level limits, transaction costs, slippage or a complete paper ledger.

9. Risk Management and Sizing

Parameter	Initial proposal	Required behavior
Virtual capital	₹5,000	Configurable ledger starting value; confirm with user.
Cash reserve	10%	Do not allocate reserve to new entries.
Maximum position value	25% of equity	Also cap by available cash and whole-share constraints.
Maximum planned risk/trade	₹25	Quantity must fit risk, position value, cash and liquidity limits.
Maximum daily loss	₹75	Track realized P&L separately from open planned risk.
Maximum trades/day	3	Document what counts as a trade.
Maximum simulated order events/day	6	Count entry and exit events consistently.
Minimum reward:risk	1.5:1 candidate	Calculate from simulated entry, stop and target.
Shorts/options/margin	Disabled	Enforce in code, not just UI/config.
Stop required	Yes	No simulated entry without a valid stop.

9.1 Quantity formula

Risk-based quantity = floor(maximum permitted rupee risk / absolute(entry − stop)). Then cap quantity by maximum position value, available cash after reserve, whole-share constraints and liquidity rules. If quantity is zero, reject the candidate.

9.2 Separate risk measures

Planned open risk: total initial risk at stops for open paper positions.

Realized daily P&L: closed simulated trades after modeled fees and slippage.

Unrealized P&L: estimated mark-to-market P&L for open positions.

Daily circuit breaker: blocks new entries after the configured limit; simulated exits remain available.

Configuration changes must be logged and attached to subsequent trades.

10. Technical Architecture

Module	Responsibility	Suggested path
Config	Validated settings and risk/strategy parameters.	nse/config.py
Universe	Dated Nifty 50 membership.	nse/universe/

Instrument master	Symbol/token/exchange mapping.	nse/data/instruments.py
Provider interface	Fixtures, CSV, history and streaming abstraction.	nse/data/provider.py
Angel One adapter	Market-data endpoints only; authentication/session handling.	nse/data/angelone.py
Candle validator/cache	Normalize, validate, deduplicate and cache bars.	nse/data/candles.py
Strategy	Pure, explainable ORB functions.	nse/strategy/orb.py
Risk engine	Quantity, exposure, stop/target and daily limits.	nse/risk/
Paper broker	Simulated fills, costs, positions and ledger only.	nse/paper/
Backtester	Chronological replay and metrics.	nse/backtest/
Persistence	Local SQLite state, trades and config snapshots.	nse/storage/
Dashboard	Local views for signals, risk, portfolio and health.	nse/ui/
Tests	Unit, integration, regression and safety tests.	tests/nse/

Architecture rule: strategy code cannot call a broker API. The data adapter supplies market data; the strategy emits a candidate; risk checks approve/block; only the paper engine simulates fills. No live broker execution module will be built.

11. Market Data and Calendar

NSE regular equity trading is generally 09:15–15:30 IST on trading days, excluding holidays and special sessions. The system must use an up-to-date calendar and account for official notices. Angel One's official SmartAPI documentation describes historical candle data with timestamp, OHLC and volume; five-minute intervals; and a documented per-request historical window. Its market-data documentation describes requests for up to 50 symbols with a stated one-request-per-second limit. These API details must be rechecked against current official docs and account entitlements at implementation time.

Record source timestamp, retrieval timestamp and last-success timestamp.

Normalize timestamps to Asia/Kolkata; handle candle-start versus candle-end conventions.

Never trigger on an incomplete candle.

Handle throttling, retry/backoff, auth expiry, empty/partial responses and network loss.

Cache historical data locally for reproducible backtests and to avoid repeated calls.

Validate corporate actions, missing bars, circuit limits and low-liquidity behavior.

Use dated constituent snapshots; do not assume today's Nifty 50 membership applied historically.

12. Dashboard Requirements

View	Contents
System overview	Market/session status, data timestamp, feed health, active config and paper-only banner.
Scanner	Symbol, signal state, trigger reason, candle timestamp, entry/stop/target estimate and volume result.
Risk panel	Virtual equity, cash, reserve, planned open risk, daily P&L, limits remaining and trade/order counts.
Paper portfolio	Positions, quantity, simulated entry, stop/target and estimated unrealized P&L.
Trade journal	Entry/exit timestamps, config version, fill assumptions, fees/slippage, realized P&L and risk verdict.
Backtest report	Net return, trade count, win rate, average win/loss, expectancy, profit factor, max drawdown and costs.
System log	Data gaps, stale-data blocks, API failures, risk blocks and exceptions.

13. Phase-by-Phase Delivery Plan

Phase	Work	Definition of done	Estimate
0 Foundation	Confirm scope, capital, environment, baseline and paper-only boundary.	Written requirements and clean project structure.	0.5–1 day
1 Scanner core	Synthetic fixtures, candle validation, ORB, volume filter, console output.	Tests cover valid/invalid candles, breakout, volume, missing range and boundaries.	1–2 days
2 Universe	Dated Nifty 50 file and verified symbol-token mapping.	Unique symbols; unknowns reported; effective date visible.	1–2 days
3 Data provider	Provider interface, local cache, SmartAPI history and optional streaming.	Retry, rate-limit, stale-data and response-validation tests pass.	2–4 days
4 Risk/paper ledger	Sizing, exposure/cash rules, daily circuit breaker, costs and SQLite ledger.	Risk tests pass; no order endpoints; paper fills are deterministic.	3–5 days
5 Backtesting	Chronological replay, costs, slippage, metrics and out-of-sample tests.	Reproducible logs, no look-ahead and ambiguity tests.	3–6 days
6 Dashboard	Local status, signals, portfolio, risk, journal and exports.	Paper banner, health warnings and exports work.	3–5 days
7 Paper observation	Run, reconcile candles/logs, resolve operational defects.	No critical defects across recommended 20–30 sessions.	20–30 trading sessions
8 Maintenance	API/constituent updates, backup, regression and controlled expansion.	Versioned releases and restore test.	Ongoing

14. Testing and Validation

Unit tests: candle validity, time boundaries, ORB range, breakout, volume threshold, stop/target, quantity rounding and risk checks.

Integration tests: provider → validator → strategy → risk engine → paper engine → ledger.

Regression tests: fixed inputs produce stable outputs unless a change is documented.

Safety tests: prove no broker order-placement method is imported or called; live mode is rejected.

Data tests: duplicate timestamps, gaps, stale bars, partial responses, malformed payloads, wrong tokens, holidays and special sessions.

Backtest tests: chronological replay, no future data in earlier decisions, consistent fees/slippage, conservative stop/target conflict handling.

Operational tests: network loss, throttling, expired authentication, restart, corrupt state, disk-full and backup restore.

Paper observation: compare timestamps/prices with provider data and log discrepancies rather than silently correcting them.

15. Performance Evaluation Standards

Do not judge a strategy by win rate alone. Report:

Number of trades and coverage period.

Gross and net P&L after modeled fees and slippage.

Win rate, average win/loss, payoff ratio and expectancy per trade.

Profit factor, maximum drawdown, largest loss, consecutive losses and time in market.

Performance by symbol, time of day, weekday and regime where sample sizes permit.

Out-of-sample results, parameter sensitivity and comparison with a baseline.

Known limitations: survivorship bias, data quality, small sample size and execution assumptions.

Backtests can overfit and differ from real execution. A positive historical result does not prove future profitability; no configuration should be described as 100% successful.

16. Security and Safety

No real order submission code, endpoint or broker order capability in the application.

Use a market-data-only integration where possible; do not request trading permissions for this project.

Never store API keys, passwords, PINs, TOTP seeds/codes or session tokens in source files, screenshots, logs, chat or Git.

Keep secrets in local environment variables or an approved secret store; add secret files to .gitignore.

Bind the dashboard to localhost by default; do not expose it publicly.

Back up configuration, database and logs.

Show paper-only banners throughout the application; no hidden live-mode switch.

Fail closed: invalid config, stale data, unknown instrument or corrupt risk state blocks new entries.

17. Risks and Mitigations

Risk	Impact	Mitigation
API terms/access change	Data may be unavailable or use restricted.	Verify official docs, account entitlement and terms; isolate adapter.
Stale/incomplete data	False or late signals.	Timestamp bars; freshness checks; block incomplete candles.
Incorrect token mapping	Wrong stock data.	Validate against official instrument master; test mapping.
Survivorship bias	Inflated backtest.	Use point-in-time constituents where available; disclose limitation.
Overfitting	Results fail out of sample.	Freeze rules, out-of-sample split, walk-forward and sensitivity tests.
Costs/slippage ignored	Overstated returns.	Configurable costs and conservative fills; stress-test assumptions.
Illiquid symbols	Unrealistic fills/impact.	Liquidity/spread filters and position caps.
Network/laptop interruption	Missing bars or state inconsistency.	Persist state, detect gaps, resync and block entries.
Scope creep into live execution	Unintended financial exposure.	No order API implementation; isolated paper engine.
Session/calendar changes	Wrong opening range or cutoff.	Use official calendar and review exchange notices.

18. Acceptance Criteria for First Complete Release

Scans a dated, validated Nifty 50 universe without duplicate symbols.

Uses validated completed five-minute candles and blocks stale/incomplete data.

Produces an explainable ORB signal or a clear reason for no signal.

Calculates entry, stop, target and quantity with explicit assumptions and correct rounding.

Enforces risk limits and logs PASS/BLOCK reasons.

Simulates fills, fees, slippage, exits, positions and P&L in a persistent local ledger.

Backtest is chronological and includes costs, drawdown, metrics and trade logs.

Dashboard shows paper mode, feed health, risk state and warnings.

Automated tests pass and a safety test confirms there is no real-order path.

User guide covers setup, configuration, operation, troubleshooting, backups and limitations.

19. Proposed Repository Layout

day-trader-skill/

nse/

config.py

universe/ (nifty50.csv, instruments.py)

data/ (provider.py, angelone.py, candles.py, cache.py, calendar.py)

strategy/ (orb.py)

risk/ (checks.py, sizing.py)

paper/ (broker.py, portfolio.py, ledger.py, costs.py)

backtest/ (engine.py, metrics.py, report.py)

storage/ (database.py)

ui/ (app.py)

tests/nse/

test_candles.py

test_orb.py
test_risk.py
test_paper.py
test_backtest.py
test_safety.py
docs/ (project-report.md, user-guide.md)
.env.example
.gitignore

20. Draft Configuration

Setting	Draft value	Notes
mode	paper_only	Enforced in code; live mode does not exist.
market	NSE cash equities	No derivatives in v1.
universe	Nifty 50	Dated official list.
timezone	Asia/Kolkata	Do not infer from PC timezone.
candle_interval	5 minutes	Only completed bars trigger signals.
virtual_capital_inr	5000	Confirm with user.
cash_buffer_pct	10	Configurable.
max_position_pct	25	Initial test parameter.
max_risk_per_trade_inr	25	Test parameter, not financial advice.
max_daily_loss_inr	75	Track realized P&L and open risk separately.
max_trades_per_day	3	Define trade counting.
max_simulated_events_per_day	6	Count entry/exit consistently.
min_reward_risk	1.5	Research candidate; test after costs.
allow_short/options/margin	FALSE	Enforce in code.

21. Decisions Needed

Confirm whether ₹5,000 is the desired virtual starting capital.

Confirm whether Angel One SmartAPI is acceptable as the proposed data source; do not share credentials.

Choose console-first development or a local web dashboard after the core tests pass.

Confirm long-only five-minute ORB remains the first candidate and options, shorts, margin and real orders remain excluded.

Set the required paper-observation period and evidence threshold before expanding the universe.

22. Official References

Official pages consulted for planning; verify current API documentation, access terms, limits and exchange notices again during implementation.

NSE India — Nifty 50 index and constituent list: https://www.nseindia.com/static/products-services/indices-nifty50-index

NSE India — Market timings: https://www.nseindia.com/static/market-data/market-timings

Angel One SmartAPI — official documentation: https://smartapi.angelone.in/docs

Angel One SmartAPI — historical API documentation: https://smartapi.angelone.in/docs/Historical

23. Final Recommendation

Proceed in small, testable phases: finish the scanner core with synthetic data and automated tests; establish the official universe and instrument mapping; add historical data; then build risk controls, the paper ledger and backtester; add the dashboard after those components are reliable. Do not add real-order functionality. Success means a reliable, explainable and auditable research tool—*not* a guarantee of profitable trades.